



ANDROID STATIC ANALYSIS REPORT

app_icon

 **Projet WS (1.0)**

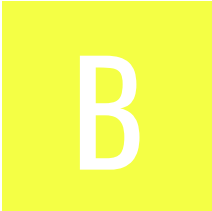
File Name:

app-debug.apk

Package Name: com.example.projets

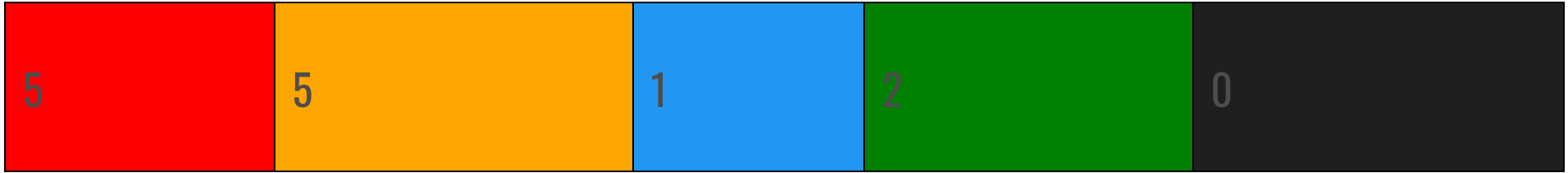
Scan Date: May 15, 2026, 3:22 p.m.

App Security Score: 40/100 (MEDIUM RISK)

Grade: 

 FINDINGS SEVERITY





FILE INFORMATION

File Name: app-debug.apk
Size: 10.62MB
MD5: 4a15523a9e071e8fcf44dc2fb20ef378
SHA1: f42c1acac2e5eb6491d7d27847c3f11cac094e6c
SHA256: d48453794f399f6d4806a94f8e31f935d7fd15154f243c74d07209c50506fc6b

APP INFORMATION

App Name: Projet WS
Package Name: com.example.projetsws
Main Activity: com.example.projetsws.ui.MainActivity
Target SDK: 36
Min SDK: 24
Max SDK:
Android Version Name: 1.0
Android Version Code: 1

APP COMPONENTS

Activities: 3
Services: 3
Receivers: 2
Providers: 2
Exported Activities: 0
Exported Services: 0
Exported Receivers: 1
Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed
v1 signature: False
v2 signature: True
v3 signature: False
v4 signature: False
X.509 Subject: CN=Android Debug, O=Android, C=US
Signature Algorithm: rsassa_pkcs1v15
Valid From: 2026-03-16 10:24:22+00:00
Valid To: 2056-03-08 10:24:22+00:00
Issuer: CN=Android Debug, O=Android, C=US
Serial Number: 0x1
Hash Algorithm: sha256
md5: faca3051ae4d779d44cd03c47300c736
sha1: 05c4202a39a2b284afe214f123055c97a63ac0cb
sha256: ebfbf1c8cd192ec8b09984bd239cc5dcb986b864e1591572f401ce0d4699eb28
sha512: 9a51e0fd6d683006d62dfd9ea804000a770e634381e6a50ef8a8fb2e97fc5473ac14498a097addbf15976da043278af1c3ac6e213fbbd9d65ee4c029abe49923
PublicKey Algorithm: rsa
Bit Size: 2048
Fingerprint: 91a1f71b3206f8f2343074a223f34d019731a2e34d87b42070d696899768d762
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.
android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all networks.
com.example.projetsws.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	unknown	Unknown permission	Unknown permission from android reference

APKID ANALYSIS

FILE	DETAILS	
classes3.dex	FINDINGS	DETAILS
	Compiler	r8
classes4.dex	FINDINGS	DETAILS
	Compiler	r8
classes5.dex	FINDINGS	DETAILS
	Compiler	r8
classes2.dex	FINDINGS	DETAILS
	Compiler	unknown (please file detection issue!)
classes7.dex	FINDINGS	DETAILS
	Compiler	r8 without marker (suspicious)

FILE	DETAILS	
classes6.dex	FINDINGS	DETAILS
	Anti-VM Code	Build.MANUFACTURER check
	Compiler	r8 without marker (suspicious)
classes.dex	FINDINGS	DETAILS
	Anti-VM Code	Build.FINGERPRINT check Build.MODEL check Build.MANUFACTURER check Build.PRODUCT check SIM operator check
	Compiler	r8

 NETWORK SECURITY

HIGH: 1 | WARNING: 0 | INFO: 0 | SECURE: 0

NO	SCOPE	SEVERITY	DESCRIPTION
1	192.168.0.162 10.0.2.2	high	Domain config is insecurely configured to permit clear text traffic to these domains in scope.

 CERTIFICATE ANALYSIS

HIGH: 1 | WARNING: 0 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application signed with debug certificate	high	Application signed with a debug certificate. Production application must not be shipped with a debug certificate.

Q MANIFEST ANALYSIS

HIGH: 3 | WARNING: 1 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 7.0, [minSdk=24]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Clear text traffic is Enabled For App[android:usesCleartextTraffic=true]	high	The app intends to use cleartext network traffic, such as cleartext HTTP, FTP stacks, DownloadManager, and MediaPlayer. The default value for apps that target API level 27 or lower is "true". Apps that target API level 28 or higher default to "false". The key reason for avoiding cleartext traffic is the lack of confidentiality, authenticity, and protections against tampering; a network attacker can eavesdrop on transmitted data and also modify it without being detected.
3	App has a Network Security Configuration[android:networkSecurityConfig=@xml/network_security_config]	info	The Network Security Configuration feature lets apps customize their network security settings in a safe, declarative configuration file without modifying app code. These settings can be configured for specific domains and for a specific app.
4	Debug Enabled For App [android:debuggable=true]	high	Debugging was enabled on the app which makes it easier for reverse engineers to hook a debugger to it. This allows dumping a stack trace and accessing debugging helper classes.

NO	ISSUE	SEVERITY	DESCRIPTION
5	Broadcast Receiver (androidx.profileinstaller.ProfileInstallReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.DUMP [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.

</> CODE ANALYSIS

HIGH: 0 | WARNING: 3 | INFO: 1 | SECURE: 1 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	Files may contain hardcoded sensitive information like usernames, passwords, keys etc.	warning	CWE: CWE-312: Cleartext Storage of Sensitive Information OWASP Top 10: M9: Reverse Engineering OWASP MASVS: MSTG-STORAGE-14	io/grpc/PersistentHashMapMappedTrie.java io/grpc/internal/DnsNameResolver.java io/grpc/internal/PickFirstLoadBalancerProvider.java io/grpc/internal/TransportFrameUtil.java io/reactivex/internal/schedulers/SchedulerPoolFactory.java
2	IP Address disclosure	warning	CWE: CWE-200: Information Exposure OWASP MASVS: MSTG-CODE-2	com/example/projetws/ui/AddEtudiant.java com/example/projetws/ui/ListEtudiant.java io/grpc/okhttp/OkHttpClientTransport.java io/grpc/okhttp/OkHttpServerTransport.java
3	The App uses an insecure Random Number Generator.	warning	CWE: CWE-330: Use of Insufficiently Random Values OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-6	io/grpc/internal/DnsNameResolver.java io/grpc/internal/ExponentialBackoffPolicy.java io/grpc/internal/PickFirstLeafLoadBalancer.java io/grpc/internal/PickFirstLoadBalancer.java io/grpc/internal/RetriableStream.java io/grpc/okhttp/OkHttpClientTransport.java io/grpc/util/OutlierDetectionLoadBalancer.java io/grpc/util/RoundRobinLoadBalancer.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
4	The App logs information. Sensitive information should never be logged.	info	CWE: CWE-532: Insertion of Sensitive Information into Log File OWASP MASVS: MSTG-STORAGE-3	com/example/projetws/ui/AddEtudiant.java com/example/projetws/ui/ListEtudiant.java io/grpc/okhttp/internal/Platform.java
5	This App uses SSL certificate pinning to detect or prevent MITM attacks in secure communication channel.	secure	OWASP MASVS: MSTG-NETWORK-4	io/grpc/okhttp/OkHttpClientChannelBuilder.java io/grpc/okhttp/OkHttpClientServerBuilder.java io/grpc/util/AdvancedTlsX509TrustManager.java

SHARED LIBRARY BINARY ANALYSIS

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
----	---------------	----	-----	-----------------	-------	-------	---------	---------	---------------------

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
1	armeabi-v7a/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
2	x86/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
3	arm64-v8a/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
4	x86_64/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
5	armeabi-v7a/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
6	x86/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
7	arm64-v8a/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
8	x86_64/libdatastore_shared_counter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION
----	------------	-------------	---------	-------------

BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00013	Read file and put it into a stream	file	io/grpc/TlsChannelCredentials.java io/grpc/TlsServerCredentials.java io/grpc/util/AdvancedTlsX509KeyManager.java io/grpc/util/AdvancedTlsX509TrustManager.java okio/Okio__JvmOkioKt.java
00162	Create InetAddress object and connecting to it	socket	io/grpc/okhttp/internal/Platform.java
00163	Create new Socket and connecting to it	socket	io/grpc/okhttp/internal/Platform.java

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	2/25	android.permission.INTERNET, android.permission.ACCESS_NETWORK_STATE
Other Common Permissions	0/44	

Malware Permissions:

Top permissions that are widely abused by known malware.

Other Common Permissions:

Permissions that are commonly abused by known malware.

! OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION
--------	----------------

🔍 DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION
github.com	ok	IP: 140.82.121.4 Country: United States of America Region: California City: San Francisco Latitude: 37.775700 Longitude: -122.395203 View: Google Map
192.168.0.162	ok	IP: 192.168.0.162 Country: - Region: - City: - Latitude: 0.000000 Longitude: 0.000000 View: Google Map

🔑 HARDCODED SECRETS

POSSIBLE SECRETS
23456789abcdefghijklmnopqrstuvwxyz
16a09e667f3bcc908b2fb1366ea957d3e3adec17512775099da2f590b0667322a

☰ SCAN LOGS

Timestamp	Event	Error
-----------	-------	-------

2026-05-15 15:22:17	Generating Hashes	OK
2026-05-15 15:22:17	Extracting APK	OK
2026-05-15 15:22:17	Unzipping	OK
2026-05-15 15:22:17	Parsing APK with androguard	OK
2026-05-15 15:22:18	Extracting APK features using aapt/aapt2	OK
2026-05-15 15:22:18	Getting Hardcoded Certificates/Keystores	OK
2026-05-15 15:22:23	Parsing AndroidManifest.xml	OK
2026-05-15 15:22:23	Extracting Manifest Data	OK
2026-05-15 15:22:23	Manifest Analysis Started	OK
2026-05-15 15:22:23	Reading Network Security config from network_security_config.xml	OK
2026-05-15 15:22:23	Parsing Network Security config	OK
2026-05-15 15:22:23	Performing Static Analysis on: Projet WS (com.example.projetws)	OK

2026-05-15 15:22:23	Fetching Details from Play Store: com.example.projets	OK
2026-05-15 15:22:23	Checking for Malware Permissions	OK
2026-05-15 15:22:23	Fetching icon path	OK
2026-05-15 15:22:23	Library Binary Analysis Started	OK
2026-05-15 15:22:23	Analyzing lib/armeabi-v7a/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing lib/x86/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing lib/arm64-v8a/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing lib/x86_64/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing apktool_out/lib/armeabi-v7a/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing apktool_out/lib/x86/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing apktool_out/lib/arm64-v8a/libdatastore_shared_counter.so	OK
2026-05-15 15:22:23	Analyzing apktool_out/lib/x86_64/libdatastore_shared_counter.so	OK

2026-05-15 15:22:23	Reading Code Signing Certificate	OK
2026-05-15 15:22:24	Running APKiD 3.0.0	OK
2026-05-15 15:22:29	Detecting Trackers	OK
2026-05-15 15:22:35	Decompiling APK to Java with JADX	OK
2026-05-15 15:23:27	Converting DEX to Smali	OK
2026-05-15 15:23:27	Code Analysis Started on - java_source	OK
2026-05-15 15:23:29	Android SBOM Analysis Completed	OK
2026-05-15 15:23:44	Android SAST Completed	OK
2026-05-15 15:23:44	Android API Analysis Started	OK
2026-05-15 15:23:51	Android API Analysis Completed	OK
2026-05-15 15:23:52	Android Permission Mapping Started	OK
2026-05-15 15:23:58	Android Permission Mapping Completed	OK

2026-05-15 15:23:59	Android Behaviour Analysis Started	OK
2026-05-15 15:24:36	Android Behaviour Analysis Completed	OK
2026-05-15 15:24:36	Extracting Emails and URLs from Source Code	OK
2026-05-15 15:24:38	Email and URL Extraction Completed	OK
2026-05-15 15:24:38	Extracting String data from APK	OK
2026-05-15 15:24:39	Extracting String data from SO	OK
2026-05-15 15:24:39	Extracting String data from Code	OK
2026-05-15 15:24:39	Extracting String values and entropies from Code	OK
2026-05-15 15:24:42	Performing Malware check on extracted domains	OK
2026-05-15 15:24:43	Saving to Database	OK

Report Generated by - MobSF v4.5.0

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

© 2026 Mobile Security Framework - MobSF | [Ajin Abraham](#) | [OpenSecurity](#).